REPORT

Hackathon Context Organized by the Unique Identification Authority of India (UIDAI) in association with NIC and MeitY, the 2026 online hackathon focused on **data-driven innovation for Aadhaar**. Participating teams received anonymized enrolment and update datasets to uncover patterns, trends, anomalies, or predictive indicators that support decision-making, system improvements, and societal insights.

Problem Statement Addressed Unlocking societal trends in Aadhaar enrolment and updates — daily enrolments are difficult to forecast accurately due to complex patterns in demographic and biometric data. Separate analysis of biometric and demographic datasets can mask systemic issues, leading to risks such as authentication failures, exclusion from services, or hidden operational gaps.

Project Objective Develop a machine learning model to **predict adult biometric enrollments (17+ years)** using same-day child enrollments (5–17 years) as a key predictor, leveraging family co-enrolment patterns. The model aims to provide accurate, state-level forecasts to aid resource planning and reveal societal behaviour trends reflected in Aadhaar data.

Approach & Methodology

1. Data Preparation

- Concatenated split CSV files for biometric and demographic datasets.
- Handled inconsistent date formats using pd.to_datetime(format='mixed', dayfirst=True, errors='coerce') to clean and parse timestamps.
- Dropped invalid dates and encoded categorical features (state, district) via LabelEncoder.
- Created date-encoded numeric feature for time-series modeling.

2. Feature Engineering & Insight

- Strong correlation observed: Child biometric enrollments (5–17) serve as a reliable same-day predictor for adult enrollments — reflecting family/group enrolment behaviour.
- Focused on aggregated district/pincode-level patterns to enable granular, state-wise analysis.

3. Modeling

- Regression task: Predict adult biometric counts (bio_age_17_plus) using child counts, location features, and time encoding.
- Models trained:
  - Linear Regression (baseline)
  - XGBoost Regressor (final model – superior performance)
- Evaluation: Actual vs. Predicted scatter plots (visualized in notebooks) showing good fit, especially with XGBoost.

4. Results & Visualization

- XGBoost captured non-linear patterns better than linear baseline.
- Scatter plots demonstrate alignment between actual and predicted adult enrollments.
- State-level showcase prepared (code supports aggregation and potential choropleth mapping with geopandas).

Key Insights

- Child enrollments are a powerful proxy for predicting adult biometric activity on the same day — highlighting family-driven patterns in Aadhaar uptake.
- Accurate daily forecasting can help UIDAI anticipate peak loads, allocate resources, and detect anomalies in enrolment behaviour.
- Separate biometric/demographic analysis risks missing interconnected societal signals; integrated modeling provides richer, more actionable insights.

Limitations & Future Work

- Relied on provided anonymized data; no real-time or external features added.
- Future: Incorporate time-series models (e.g., Prophet, LSTM), add geospatial visualization for district-level heatmaps, and explore anomaly detection for fraud/risk signals.

Conclusion

This project demonstrates how machine learning can unlock practical value from anonymized Aadhaar data — enabling better forecasting of adult biometric enrollments and supporting UIDAI's mission of inclusive, efficient identity services.

We thank UIDAI, NIC, and MeitY for organizing this hackathon and providing the datasets. Ready for presentation/discussion!

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